

Pap1 (SCHPO, UniProt Q01663) — Direct Nucleosome-Binding Hypothesis Review

Hypothesis under evaluation: *Schizosaccharomyces pombe* Pap1 directly binds nucleosomes as a biologically meaningful molecular function, rather than nucleosome association being an incidental signal in the 2025 transcription-factor interaction atlas.

Focus type: free_text · **Slug:** direct-nucleosome-binding · **Iteration:** 1 of 3

Executive Judgment

Verdict: Partially supported / weakly supported (do not upgrade to a confident direct-MF claim).

The hypothesis contains two separable claims that resolve differently:

1. **"The nucleosome signal is not merely incidental atlas background."** → **Supported.** In the source study (Skribbe et al., *Molecular Cell* 2025, PMID **40015273**), core histones were pervasive contaminants co-purifying in *all* low-stringency IPs, including the untagged control. Critically, under **high-stringency** conditions the histone association was retained for only three transcription factors — **Pap1, Atf1, and Pcr1** — out of the 89-TF library. Because the authors explicitly controlled for the generic histone background and Pap1 still scored, the signal is *specific*, not indiscriminate atlas noise.
2. **"Pap1 directly binds nucleosomes as a biologically meaningful molecular function (pioneering activity)."** → **Weakly supported / unresolved.** This is an *inference*, not a demonstrated activity. The evidence is co-immunoprecipitation/IP-MS, which cannot distinguish direct core-histone contact from a stable indirect/bridged association (e.g., Pap1 bound to its target DNA within a nucleosomal/chromatin context, carrying histones along). No reconstituted- nucleosome binding assay (EMSA, fluorescence anisotropy, NCAP-SELEX, Pioneer-seq) or functional nucleosome-displacement/pioneering assay has been performed for Pap1. The atlas authors themselves use suggestive language: the co-purification *"indicates interactions with the nucleosome and suggests pioneering activity."*

Most important caveat: Pap1's dominant, experimentally established molecular function is sequence-specific promoter DNA binding as a bZIP (AP-1/Jun-Fos-like) oxidative-stress activator. The nucleosome association must be interpreted against this backdrop; "direct nucleosome binding" is currently a plausible-but-unproven secondary property carried largely by orthology analogy to human Jun/Fos.

Evidence Matrix

#	Citation	Evidence type	Stance	Claim tested	Key finding	Context	Con- lim
1	Skribbe et al. 2025, P 40015273 (Mol Cell)	Interaction (IP-MS)	Qualifies / partially supports	Pap1 physically associates with nucleosomes	Under high- stringency IP- MS, histone co- purification is uniquely retained for Pap1, Atf1, Pcr1 (vs. pervasive low- stringency background incl. untagged control)	<i>S. pombe</i> , endogenously tagged TFs, IP-MS	Sig spe co-I bin can reso dire brid in v nuc assa
2	Skribbe et al. 2025, P 40015273	Review/ interpretation within primary paper	Qualifies	Pioneering activity	Authors state co-purification "suggests <i>pioneering activity</i> "; note Pap1 "has not been linked to nucleosome remodeling"	<i>S. pombe</i>	Exp infe no fun pion assa pom
3	UniProt/ PomBase Q01663 (GO:0031491, EXP)	Database	Supports (annotation exists)	Nucleosome binding is annotated	GO:0031491 "nucleosome binding" now present with EXP evidence, traceable to the atlas co- purification	Curated record	EXP ove co- pur pro is a inte stud
4	UniProt Q01663 feature table	Structural/ domain	Competing (primary MF)	Pap1's core molecular function	bZIP domain (76–139), basic DNA-contact motif (81–102), leucine zipper	Sequence/ domain	Eta can seq spe DNA

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					(104–111); redox CRDs; NES/NLS		TF a prin
5	PomBase GO MF (IDA): GO:0001228, GO:0000978, GO:0008301	Direct assay / curated	Competing (primary MF)	Sequence- specific promoter DNA binding	Pap1 has IDA- level activator activity, Pol II cis-regulatory sequence- specific DNA binding, and DNA bending	<i>S. pombe</i>	Wel sup don fun
6	Fellas et al. 2026, P 41326690	Mutant phenotype / localization	Competing (primary MF)	Pap1 acts via promoter occupancy	"Pap1 nuclear localisation and recruitment to promoters of oxidoreductase and membrane transporter genes is increased," driving efflux	<i>S. pombe</i> epimutants	Rein seq spe pro bin me
7	Lai et al. 2021, P 34740252	Mutant phenotype / occupancy	Competing / context	Pap1 promoter occupancy is chromatin- context dependent	Histone variant replacement (H2A.Z) <i>increases</i> Pap1 occupancy at drug- resistance promoters	<i>S. pombe</i> + AtOXS3	Sug Pap is fa by chr rem — a aga stro intr pion
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#	Citation	Evidence type	Stance	Claim tested	Key finding	Context	Con- lim
	Human ortholog literature (Jun/Fos) cited by P 40015273	Structural/ evolutionary	Supports (by analogy)	bZIP factors can bind nucleosomes	Jun/Fos bind nucleosomes in vitro and associate with nucleosome-rich regions in vivo	Human/in vitro	Orti ana only dire evid pom
9	Convergent bZIP pattern in atlas (Atf1, Pcr1 also retain histones; Zip1 does not)	Structural/ evolutionary	Supports (plausibility)	Nucleosome binding may be a bZIP-family trait	Three of four mitotic bZIPs co-purify histones; authors propose family-wide property	<i>S. pombe</i>	Cor con with dire bin sha chr asse beh

GO Curation Implications (leads requiring curator verification)

- **Term in question:** [GO:0031491 nucleosome binding](#) (MF), currently on Q01663/PomBase with evidence code **EXP**.
- **Assessment:** The evidence type (high-stringency co-IP/IP-MS) supports a **physical interaction with histones/nucleosomes**, but an **EXP** code on a *direct* MF term is arguably **too strong** for what is a co-purification that cannot resolve direct vs. bridged binding.
- **Recommended lead actions (curator to verify):**
- **Retain but re-scope the evidence, not remove the term.** The specific high-stringency signal justifies keeping a nucleosome/histone-interaction annotation rather than deleting it.
- **Prefer an interaction-flavored provenance.** Consider annotating with evidence code **IPI** (physical interaction) with the histone/nucleosome as the "with" object, or downgrading the directness implied by EXP, so the annotation reflects "co-purifies with nucleosome" rather than a validated direct binding activity.

- **Consider** **GO:0042393 histone binding** as an equally or more *literal* description of what was observed (co-purification of **core histone subunits**), pending a discriminating assay. **GO:0031491 nucleosome binding** is the atlas's chosen framing and is defensible, but curators should note it is model-dependent.
- **Mark as non-core / tentative.** Pap1's core MF annotations remain **GO:0001228**, **GO:0000978**, **GO:0008301**. Nucleosome binding should be treated as a **secondary, provisional** function, ideally with a curator note that it rests on a single interaction study and orthology.
- **Do not** promote to functional pioneer-factor process/MF terms — no pioneering/remodeling activity has been demonstrated for Pap1.

(Per instructions, "protein binding" is deliberately avoided as the recommendation; "histone binding" / "nucleosome binding" with interaction-level evidence is the more informative option.)

GO decision table (lead — curator to verify)

GO term	Aspect	Current status	Evidence basis	Recommended lead action
GO:0031491 nucleosome binding	MF	Present, EXP (Q01663/PomBase)	Single high-stringency IP-MS co-purification (P 40015273); no direct/in vitro assay	Retain but re-scope: change evidence from EXP→ IPI (with = histones/nucleosome); mark non-core / provisional ; add note that directness & pioneering are inferred
GO:0042393 histone binding	MF	Not annotated	More literal description of "co-purifies with core histone subunits"	Candidate alternative to GO:0031491 if curator prefers matching term to the actual observable
GO:0001228 DNA-binding transcription activator activity, Pol II-specific	MF	Present, IDA	Multiple primary studies	Retain — core function (unchanged)
GO:0000978 RNA Pol II cis-regulatory region sequence-specific DNA binding	MF	Present, IDA	Multiple primary studies	Retain — core function (unchanged)
GO:0008301 DNA binding, bending	MF	Present, IDA	Primary	Retain (unchanged)
(pioneer/remodeling MF or BP terms)	MF/BP	Not annotated	None (no functional pioneering assay for Pap1)	Do NOT add — unsupported

Mechanistic Scope

- **Immediate molecular activity tested:** whether Pap1 makes a *direct physical contact with the nucleosome core particle* (histone octamer $\pm$ nucleosomal DNA) as an intrinsic binding activity.
- **What the data actually show:** Pap1 **co-purifies** with core histones under stringent IP conditions — a physical-association readout at the level of the assembled complex, not a reconstituted binary binding measurement.
- **Separated downstream/secondary layers (NOT the direct activity):**
 - Transcriptional activation of oxidative-stress/drug-resistance regulons (BP consequence).
 - Nuclear accumulation via redox-controlled Crm1/NES switching (regulatory input, not nucleosome MF).
 - Increased efflux and drug resistance phenotypes (organismal/phenotypic downstream effects).
 - Chromatin-context effects on Pap1 occupancy (e.g., H2A.Z replacement *enabling* Pap1 access,

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— this frames Pap1 as a *beneficiary* of remodeling rather than a demonstrated pioneer that opens chromatin itself.

Conflicts and Alternatives

1. **Direct vs. bridged interaction (primary alternative).** IP-MS cannot distinguish Pap1 $\leftrightarrow$ histone direct contact from Pap1 bound to nucleosomal target DNA that co-precipitates histones. Pap1's very strong, well-documented sequence-specific DNA binding makes the "carried along on chromatin" explanation a serious competitor. High stringency reduces but does not eliminate this ambiguity.
2. **Orthology overreach.** The strongest positive framing (Jun/Fos bind nucleosomes in vitro) is an analogy across a >500 Myr divergence; pombe Pap1 has diverged CRDs and its own regulatory logic. Orthology motivates the hypothesis but is not evidence of the pombe activity.

3. Chromatin facilitates Pap1, not vice versa.

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shows histone-variant replacement *increases* Pap1 promoter occupancy — consistent with Pap1 needing accessible chromatin rather than acting as a strong intrinsic nucleosome invader/pioneer.

4. **Shared-artifact possibility across bZIPs.** Atf1/Pcr1 (an obligate heterodimer) and Pap1 all scoring could reflect a common bona fide bZIP property *or* a shared behavior (e.g., particularly stable chromatin residence) that co-retains histones under stringency without direct core contact.
 5. **Database carry-over risk.** Once GO:0031491 (EXP) is on the record, it can propagate to orthologs/paralogs by inference; curators should ensure downstream IBA/ISS annotations do not over-generalize an unvalidated pioneering activity.
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Knowledge Gaps

Gap	What was checked	Why it matters for curation	What would resolve it
Direct vs. indirect binding	Confirmed evidence is IP-MS co-purification (P 40015273); no in vitro assay found	Determines whether GO:0031491 EXP (direct MF) is warranted vs. an interaction annotation	Reconstituted-nucleosome EMSA / fluorescence anisotropy with recombinant Pap1 (or its bZIP domain)
Functional pioneering	No pombe nucleosome-displacement/ accessibility data for Pap1	"Biologically meaningful function" implies a functional consequence, not just contact	ATAC/MNase accessibility change at Pap1 sites upon induction; ISO/Pioneer-seq assay
DNA-dependence of the co-IP	Not tested in source	If benzonase/DNase abolishes the histone co-IP, the interaction is DNA-bridged, not direct	Nuclease-treated IP-MS; salt-titration of the Pap1-histone co-complex
Which histones / stoichiometry	Atlas reports "core histone subunits" generically	Distinguishes octamer/ nucleosome binding from single-histone chaperone-like binding	Quantitative crosslinking-MS; defined-nucleosome pulldowns
bZIP-family generality	Zip1 negative; Atf21/ Atf31 untested (meiotic)	Determines whether this is a conserved MF worth family-wide annotation	Meiotic-cell IP-MS of Atf21/Atf31; comparative in vitro binding

Discriminating Tests (recommended, ranked)

- 1. Reconstituted-nucleosome binding assay** (EMSA or fluorescence anisotropy) with recombinant full-length Pap1 and its isolated bZIP domain on Widom-601 nucleosomes ± cognate AP-1 motif. *Directly tests "direct binding"; discriminates sequence-directed vs. core-histone binding.*

2. **Nuclease-controlled re-IP-MS:** repeat the Pap1 pulldown $\pm$ benzonase and $\pm$ high salt. *If histone co-purification is DNA-dependent $\rightarrow$ association is chromatin-bridged, not a direct nucleosome MF.*
3. **Pioneer-seq / NCAP-SELEX** with Pap1 to quantify nucleosomal vs. free-DNA binding preference and end vs. dyad positioning.
4. **In vivo accessibility change:** ATAC-seq/MNase-seq before and after Pap1 activation (H₂O₂) at Pap1 target loci, in *pap1 Δ* vs. wild type. *Tests functional pioneering.*
5. **Comparative bZIP panel** (Atf1, Pcr1, Zip1, meiotic Atf21/Atf31) in the same in vitro assay to test whether nucleosome binding is a genuine family trait.

Curation Leads (require curator verification)

- **Candidate reference to attach/verify:** Skribbe M, Soneson C, Stadler MB, Schwaiger M, et al. *A comprehensive Schizosaccharomyces pombe atlas of physical transcription factor interactions with proteins and chromatin*. Mol Cell. 2025;85(7):1426–1444.e8.

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- Snippet to verify in full text: histone co-purification "uniquely preserved for Pap1 and the heterodimeric TFs Atf1 and Pcr1 ... under high stringency conditions," and this "indicates interactions with the nucleosome and **suggests** pioneering activity." (Verify verbatim; the published abstract does not itself mention Pap1/nucleosome — the claim is in the Results/Discussion.)
- **Candidate GO action:** Retain `G0:0031491 nucleosome binding` **but** re-scope evidence from EXP to an interaction-level code (IPI, "with" = histones/nucleosome), OR replace with `G0:0042393 histone binding`; annotate as **non-core / provisional** with a curator note.
- **Candidate qualifier:** Add note that directness and pioneering are **inferred**, based on a single co-purification study plus Jun/Fos orthology.
- **Suggested curator questions:**
 - Does the atlas supplement report DNase/benzonase-resistance of the Pap1–histone co-IP?
 - Is there any reconstituted-nucleosome or accessibility assay for Pap1 (none found here)?
 - Should nucleosome binding for Atf1/Pcr1 be curated consistently with Pap1?
- **Suggested experiments:** the five discriminating tests above (priority: #1 and #2).

Computed Provenance: bZIP basic-region conservation (Iteration 2)

Sequences fetched from UniProt (Q01663, P52890, Q09926, Q9Y7P6, P01100, P05412); ungapped positional comparison of the basic-region→leucine-zipper window (28 aa) against Pap1.

Protein	bZIP basic-region window (basic region → leucine zipper)	% identity vs Pap1	AP-1 redox Cys in basic region
Pap1 (Q01663)	RKAQNRAAQRAFRKRKEDHLKALETQVVTLKE	100% (ref)	No
Atf1 (P52890)	FLERNRQAALKCRQRKKQWLSNLQAKVEFYGN	32.1%	Yes
Pcr1 (Q09926)	ILERNRIAASKFRQKKKEWIKLEQTANAAFE	32.1%	No
FOS (P01100)	RRERNKMAAAKCRNRRRELTDTLQAETDQLED	21.4%	Yes
JUN (P05412)	KRMNRRIAASKCRKRKLERIARLEEKVKTLKA	35.7%	Yes
Zip1 (Q9Y7P6)	(basic region not auto-detected by heuristic)	—	—

Interpretation (conservative): - Pap1 has a *bona fide* bZIP basic region carrying the invariant **N-R-A-A-Q-R** signature, so the Jun/Fos orthology is real **at the diverged-fold level** — but sequence identity to human JUN/FOS is low (21–36%), so the "pioneer like Jun/Fos" argument is weak analogy, not homology-grade evidence. - Pap1 **lacks** the AP-1 basic-region redox cysteine present in FOS/JUN/Atf1; Pap1 instead senses redox via separate n-CRD/c-CRD domains — a documented mechanistic divergence from the AP-1 paradigm. - In Jun/Fos the **same** basic-region helix reads free DNA *and* nucleosomal DNA. Thus, if Pap1 engages nucleosomes, the fold predicts **nucleosomal-DNA reading via its DNA-binding basic region**, i.e., an extension of sequence-specific DNA binding onto chromatin — *not* a separate histone-core protein-protein MF. This nuance should shape the GO term choice (nucleosomal-DNA/chromatin binding vs. a distinct nucleosome/histone-binding MF) and argues further against an unqualified, high-confidence "direct nucleosome binding" MF call.

Note: Attempts to retrieve the atlas full text (bioRxiv / Cell) for the decisive methods detail — whether the Pap1–histone co-IP is nuclease/DNase-resistant — were blocked (HTTP 429/403). That single fact (DNA-dependence of the co-purification) remains the key unresolved discriminator and should be checked by a curator in the paper's supplement.

Limitations of this review

- Programmatic access: PubMed and UniProt REST were queried successfully; the bioRxiv/Cell full text was rate-limited (HTTP 429), so the exact Results-section wording was obtained via web search summaries and should be verified verbatim by a curator before quoting in the annotation.
- The published abstract of

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does not mention Pap1 or nucleosomes; the finding lives in the body, limiting snippet-level validation.

- No local `*-bioinformatics` analyses were available (intentionally withheld); conclusions rest on primary literature, the public atlas, and UniProt/PomBase records.
-